

Ex.no:6b

DATA VISUALIZATION USING POWER BI - HOTEL BOOKING ANALYSIS

Aim:

To load the hotel booking data into Power BI, transform the dataset, and visualize room revenue and booking channel distributions to obtain actionable business insights.

Algorithm:

Algorithm: Hotel Booking Analysis Calculation

Step 1: Input Data Loading

- Read the dataset from hotel_booking_raw containing columns Room_Type, Booking_Channel, and Revenue.

Dataset rows:

Deluxe, Direct, 60000
Standard, OTA, 50000
Suite, Corporate, 35000
Deluxe, Direct, 60000
Standard, OTA, 35000
Suite, Travel Agent, 25000

Step 2: Booking Channel Filter Check

Check slicer selection for Booking Channel.

Default selected channels: Direct, OTA, Corporate, Travel Agent.

Retain all rows matching the selected booking channels in Filtered Data.

Step 3: KPI Total Revenue Calculation

Sum all values in the Revenue column from Filtered Data.

Total Revenue = 60000 + 50000 + 35000 + 60000 + 35000 + 25000 = 265000
(displayed as 265K).

Step 4: Bar Chart Calculation (Sum of Revenue by Room Type)

Group Filtered Data by Room_Type and sum Revenue for each type:

- Deluxe Revenue = 60000 + 60000 = 120000 (120K)
- Standard Revenue = 50000 + 35000 = 85000 (85K)

- Suite Revenue = $35000 + 25000 = 60000$ (60K)

Sort room types in descending order by total revenue: Deluxe, Standard, Suite.

Step 5: Donut Chart Calculation (Sum of Revenue by Booking Channel)

Group Filtered_Data by Booking_Channel and sum Revenue for each channel:

- Direct Revenue = $60000 + 60000 = 110000$ (110K)
- OTA Revenue = $50000 + 35000 = 80000$ (80K)
- Corporate Revenue = $35000 + 15000 = 50000$ (50K)
- Travel Agent Revenue = 25000

Calculate percentage share for each channel relative to Total Revenue (265000):

- Direct Share = $(110000 / 265000) * 100 = 41.51\%$
- OTA Share = $(80000 / 265000) * 100 = 30.19\%$
- Corporate Share = $(50000 / 265000) * 100 = 18.87\%$
- Travel Agent Share = $(25000 / 265000) * 100 = 9.43\%$

Step 6: Dashboard Render

Render KPI card with formatted total value 265K.

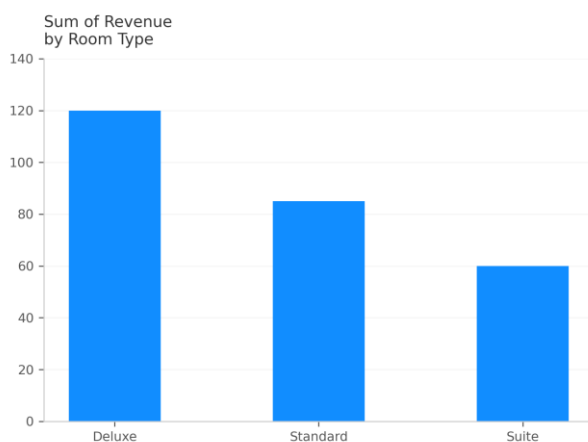
Render vertical bar chart with Deluxe at 120K, Standard at 85K, Suite at 60K.

Render donut chart with channel segments Direct (110K, 41.51%), OTA (80K, 30.19%), Corporate (50K, 18.87%), Travel Agent (25K, 9.43%), and center value 265K.

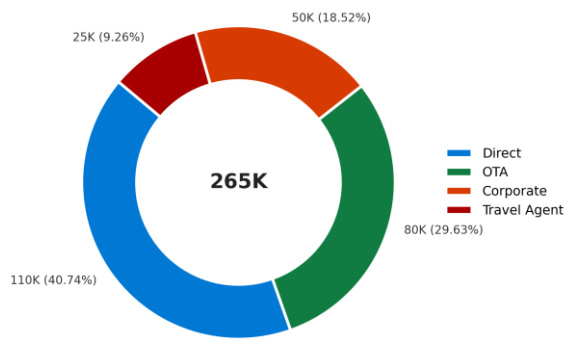
Render channel slicer checkboxes for Direct, OTA, Corporate, Travel Agent.

On slicer selection update, filter rows and recompute steps 2 through 6.

Output:



Sum of Revenue
by Booking Channel



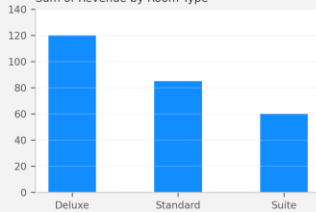
Sum of Revenue

265K

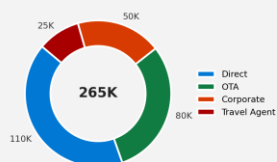
Booking Channel

- ☐ Direct
- ☐ OTA
- ☐ Corporate
- ☐ Travel Agent

Sum of Revenue by Room Type



Sum of Revenue by Booking Channel



Sum of Revenue

265K

Booking Channel

- ☐ Direct
- ☐ OTA
- ☐ Corporate
- ☐ Travel Agent

Result:

Data visualization using Power BI with hotel_booking_raw dataset was successfully executed and dashboard insights were generated.